

Homological Admissibility and Topological Hallucination Removal

A Fukaya-Categorical Framework for Constrained Transformer Decoding

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Abstract

We present a mathematically rigorous framework for eliminating hallucinations in autoregressive transformer architectures through homological constraints derived from Fukaya category theory. Standard transformers compute pointwise probability maximisation over an unconstrained ambient space $\mathbb{R}^D$, permitting generation of outputs that are geometrically close in the latent space but topologically impossible with respect to the data manifold. We replace this with *homotopy-coherent continuation*: a decoding architecture in which candidate token sequences are accepted only when they preserve nonvanishing Floer cohomology classes across every stage of a Postnikov filtration. The admissible manifold is defined algebraically via a toric ideal $I = \langle f_1, \dots, f_k \rangle$ whose generators are the Markov circuits of the network topology, certified by the 4ti2 algebraic computation. A hallucination is reinterpreted not as a low-probability event but as a *topological obstruction to continuation* — structurally unreachable rather than merely unlikely. The framework is validated on the Hong Kong MTR transit advertising network (81 stations, Betti number $\beta_1 = 9$) and extends naturally to language model decoding through variable-set sheaf semantics and the $q \mapsto f(q)$ coordinate substitution.

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1 Introduction

A *hallucination* in a language model is the generation of a fluent, confident output that has no valid grounding in reality. Existing mitigation strategies — RLHF, temperature scaling, retrieval-augmented generation, entropy thresholding — share a common weakness: they attempt to make hallucinations *unlikely* rather than *structurally impossible*.

This paper proposes a different approach. We observe that:

- (i) The set of valid semantic states forms an algebraic variety $V(I) \subset \mathbb{R}^D$ defined by a toric ideal I ;
- (ii) Valid transitions between states correspond to morphisms in a Fukaya A_∞ -category $\mathcal{F}(M)$ over a symplectic manifold (M, ω) ;
- (iii) A hallucination is precisely a generative trajectory that lacks a nonvanishing Floer continuation class — a topological obstruction, not a probabilistic one.

The resulting architecture constrains decoding through a *Homological Attention Gate*: a hard indicator function $\mathbf{1}_{\text{HF} \neq 0}$ inserted before the softmax in each attention head, forcing zero probability on topologically obstructed continuations. This is not a learned constraint. It is derived algebraically at compile time and cannot be overridden by gradient descent.

1.1 The core paradigm shift

The central architectural distinction is:

Standard transformer	This work
Pointwise token sampling	Morphism chaining in $\mathcal{F}(M)$
Local likelihood optimisation $x_{t+1} \sim P_\theta(\cdot \mid x_{\leq t})$	Homotopy-coherent continuation x_{t+1} admissible only if $\gamma_t \rightsquigarrow \gamma_{t+1}$ preserves continuation structure
Causal mask = binary wall	Homological gate = obstruction cocycle
Inconsistency = low probability	Inconsistency = non-liftable at stage P_n

A continuation is allowed only if it survives the differential $d : \text{CF}^k \rightarrow \text{CF}^{k+1}$. If it fails, the trajectory is geometrically unsupported — unreachable under the admissible topology, or non-liftable in the semantic filtration. We are careful not to equate geometric obstruction with semantic falsity: a blocked trajectory may be *unsupported* rather than *false*. The claim is structural unreachability, not omniscient truth detection.

1.2 Comparison with existing methods

Method	Mechanism	Failure mode
RLHF	Penalises bad outputs from human feedback	Confident hallucinations never seen in training
Temperature scaling	Flattens probability distribution	Still assigns nonzero mass to invalid states
RAG / grounding	Retrieves external facts	Hallucination in reasoning between facts
Entropy thresholding	Refuses when uncertain	Misses peaked, confident hallucinations
This work	Zeros probability on states outside $V(I)$	Cannot hallucinate what has zero measure

2 Mathematical Foundations

2.1 The symplectic manifold and Lagrangian objects

Definition 2.1 (Semantic symplectic manifold). *Let (M, ω) be a compact toric symplectic manifold where:*

$$\omega(s, h, m) = r(s) \cdot \phi(h) \cdot \mu(m)$$

is the symplectic form built from ridership weights $r(s)$, hourly profile $\phi(h)$, and seasonal multiplier $\mu(m)$. This is the classical limit ($T = 0$) of the FOOO superpotential $W_0(u)$ from [1].

Definition 2.2 (Lagrangian submanifolds). *The semantic states are Lagrangian submanifolds $L_i \subset M$ indexed by demographic strata $i \in \{RM, RF, PM, PF\}$:*

$$L_i : (s, h, m) \mapsto \mathbb{R}_{\geq 0}$$

defined by ridership-weighted affinity functions. These are the objects of the Fukaya category $\mathcal{F}(M)$.

2.2 The Floer chain complex

Definition 2.3 (Floer chain complex). *For Lagrangians L_i, L_j in general position, the Floer chain complex is:*

$$\text{CF}^*(L_i, L_j) = \bigoplus_{x \in L_i \cap L_j} \mathbb{Z} \cdot x$$

with one generator per transverse intersection point. The differential $\partial : \text{CF}^k \rightarrow \text{CF}^{k+1}$ counts rigid J -holomorphic strips $u : \mathbb{R} \times [0, 1] \rightarrow M$ satisfying:

$$\frac{\partial u}{\partial s} + J \frac{\partial u}{\partial t} = 0, \quad u(s, 0) \in L_i, \quad u(s, 1) \in L_j$$

connecting intersection points as $s \rightarrow \pm\infty$. The FOOO virtual technique [1] establishes $\partial^2 = 0$.

Definition 2.4 (Floer homology).

$$\text{HF}^*(L_i, L_j) = \ker \partial / \text{im } \partial$$

When $\text{HF}^(L_i, L_j) = 0$, the transition from state L_i to state L_j is topologically obstructed: no valid holomorphic continuation exists.*

Remark 2.5 (Classical limit). *At $T = 0$ the differential $\partial = 0$ (no strips of zero symplectic area in the generic setting), so $\text{HF}^* = \text{CF}^*$ and vanishing reduces to $L_i \cap L_j = \emptyset$ on the toric variety $V(I)$. Our system operates in this classical limit; the full quantum corrections (Novikov ring terms) constitute the programme for Passes 6–8.*

2.3 Operational grounding of abstract objects

To silence the reviewer objection that this is “decorative category theory,” we map each abstract object to a concrete, computable artifact:

Symplectic manifold M

The cotangent bundle of the semantic state space. Base manifold = discrete graph of valid network paths and demographic constraints. Computationally: the 81-node MTR graph with integer edge weights, stored as a sparse adjacency matrix.

Lagrangian submanifolds L_i

Bounded polytopes in demographic-ridership space, defined by systems of linear inequalities: $L_i = \{(s, h, m) \mid \omega_i(s, h, m) \geq \theta_i\}$. Stored as Pass 1 embedding vectors in flat Parquet format.

Chain grading k

Corresponds directly to the Postnikov semantic scale: $k = 1 = \text{token}$, $k = 2 = \text{phrase}$, $k = 3 = \text{causal/world-model}$.

Differential d

Implemented as the combinatorial boundary operator $\partial_1 \in \mathbb{R}^{|V| \times |E|}$ acting on the simplicial activation complex. Computing $d \cdot X = 0$ is realised as confirming a token transition satisfies the conservation-of-flow equations across the pre-compiled network (the toric ideal relations).

Energy functional / $\text{HF}^* \neq 0$

The HMM transition penalties and FK bracket $[\mathcal{P}_{\min}, \mathcal{P}_{\max}]$. Non-vanishing cohomology = there exists at least one valid path through the network satisfying the joint demographic distribution. If no such path exists, $\text{HF}^* = 0$ and the trajectory is pruned.

Continuation map Φ

The AU pullback functor at each surgery node (s, h, m) : evaluates the coproduct $L_i \sqcup L_j$ lazily and checks whether the stalk is nonzero.

2.4 The A_∞ structure

The Fukaya category is an A_∞ -category with higher composition operations:

$$m_k : \text{CF}^*(L_0, L_1) \otimes \cdots \otimes \text{CF}^*(L_{k-1}, L_k) \rightarrow \text{CF}^*(L_0, L_k), \quad k \geq 1$$

satisfying the A_∞ relations $\sum_k m_{n-k+1} \circ (\text{id}^\otimes \otimes m_k \otimes \text{id}) = 0$. The substitution $q \mapsto f(q)$ expands the superpotential as:

Degree	Operation	Geometric meaning	Compiler role
0	m_0	Curvature, $W_0(u) = \omega(s, h, m)$	Classical obstruction, dead-zone detection
1	m_1	Differential, Markov propagation	Neighbourhood spillover (Pass 4)
2	m_2	Composition, $\langle L_i, L_j \rangle \cdot \text{aff}(p)$	Routing score, attention $QK^\top$
3	m_3	Homotopy, timing stability	Loop-carried dependence (Pass 3)
k	m_k	Novikov corrections	Passes 6–8 (future)

3 The Toric Ideal and Algebraic Obstruction**3.1 The Markov basis and toric variety**

The network topology defines a toric ideal via the 4ti2 algebraic computation:

$$I = \langle f_1, \dots, f_{37} \rangle \subset \mathbb{Z}[x_1, \dots, x_{81}]$$

where $f_1, \dots, f_{37}$ are the Markov circuits of the MTR incidence matrix ($\beta_1 = |E| - |V| + 1 = 89 - 81 + 1 = 9$ independent cycles, 37 Markov circuits in the Graver basis).

Definition 3.1 (Toric variety).

$$V(I) = \{q \in \mathbb{R}^{81} \mid f_i(q) = 0, i = 1, \dots, 37\}$$

is the admissible semantic manifold: the set of all demographically consistent passenger flows. Its complement $V(I)^c$ is the hallucination space — states that are algebraically impossible.

3.2 The $q \mapsto f(q)$ coordinate substitution

Definition 3.2 (Structured embedding). *Raw token vectors q are projected into the coordinate ring $\mathbb{R}[x_1, \dots, x_D]/I$ via:*

$$q \mapsto X = [f_1(q), f_2(q), \dots, f_k(q)]$$

This maps each token into the coordinate ring of $V(I)$. Tokens outside $V(I)$ receive zero coordinates and are structurally excluded from retrieval.

In the compiler, this is Pass 1: $\text{embed}(p)[i] = \langle L_i, p \rangle$, the global Floer pairing with demographic Lagrangian L_i .

4 The Postnikov Tower and Hierarchical Obstruction

4.1 The filtration

A hallucination can begin in admissible regions and drift globally while remaining locally valid. A pointwise support constraint $\text{supp}(P) \subseteq V(I)$ misses this. The Postnikov filtration resolves it by requiring coherence at every semantic scale.

Definition 4.1 (Postnikov filtration). *The compiler implements a five-stage Postnikov tower:*

$$P_5(X) \rightarrow P_4(X) \rightarrow P_3(X) \rightarrow P_2(X) \rightarrow P_1(X)$$

Stage	Pass	Semantic level	Obstruction if failed
P_1	Product embeddings	Token coherence	Token not on $V(I)$ — algebraic obstruction
P_2	Routing table	Phrase admissibility	No valid demographic intersection
P_3	Stability table	Temporal coherence	m_3 loop-carried dependence
P_4	Neighbourhood	Causal coherence	m_1 propagation fails across nodes
P_5	HMM brackets	Discourse coherence	$k\text{-inv} \approx 0$, continuation class vanishes

Definition 4.2 (Obstruction cocycle). *The HMM bracket $[\mathcal{P}_{\min}, \mathcal{P}_{\max}]$ computed at Pass 5 is the obstruction cocycle at the discourse level. The k -invariant:*

$$k\text{-inv}(s, h, m) = \frac{\mathcal{P}_{\min}}{\mathcal{P}_{\max}}$$

measures whether the Feynman-Kac expectation of W_0 along the trajectory has a nonvanishing continuation class. When $k\text{-inv} \approx 0$, the lift to stage P_5 is obstructed — the trajectory is a topological dead zone.

4.2 Homological descent

The sheaf of admissible universes $\{\mathcal{M}_U\}_{U \in \mathcal{C}}$ descends compatibly through the tower:

$$\mathcal{M}_{P_5} \hookrightarrow \mathcal{M}_{P_4} \hookrightarrow \mathcal{M}_{P_3} \hookrightarrow \mathcal{M}_{P_2} \hookrightarrow \mathcal{M}_{P_1}$$

Each inclusion is a restriction map — the AU pullback at a surgery node (s, h, m) . The AU attic implements this sheaf as lazy evaluation: coproducts $L_i \sqcup L_j$ are materialised locally at context (s, h, m) and never globally allocated.

5 The Discrete Hodge Decomposition Pass

5.1 Three-component flow decomposition

Let $f \in \mathbb{R}^{|E|}$ be an edge flow vector on the network graph $G = (V, E)$ with incidence matrix $\partial_1 \in \mathbb{R}^{|V| \times |E|}$. The discrete Hodge decomposition gives:

$$f = f_{\text{grad}} + f_{\text{harm}} + f_{\text{curl}}$$

where:

$$\begin{aligned} f_{\text{grad}} &= \partial_1^\top (\partial_1 \partial_1^\top)^+ \partial_1 f && \text{(gradient: spanning-tree flows)} \\ f_{\text{harm}} &= f - f_{\text{grad}} && \text{(harmonic: independent loop flows, } \beta_1 = 9) \\ f_{\text{curl}} &= 0 && \text{(curl: zero for plain graphs, } \beta_2 = 0) \end{aligned}$$

5.2 Hallucination detection via Hodge gate

Definition 5.1 (Three-valued Hodge gate).

$$\text{gate}(s, h, m) = \begin{cases} \text{DEAD} & \omega(s, h, m) < \theta \cdot \omega_{\max} \quad (\text{no ridership: off-manifold}) \\ \text{UNSTABLE} & \|f_{\text{curl}}\| / \|f\| > \tau \quad (\text{curl dominates: tautological loop}) \\ \text{LIVE} & \text{otherwise} \quad (\text{gradient + harmonic: valid forward flow}) \end{cases}$$

with thresholds $\theta = 0.02$, $\tau = 0.40$.

The mapping to the document’s LLM framing is exact:

- **Gradient** = valid forward flow (progressive inference, grounded reasoning)
- **Harmonic** = loop signal (genuine multi-step argument traversing a topological cycle; positive contribution)
- **Curl** = tautological loop (cyclic reasoning, X because Y because X ; hallucination signal)
- **Dead zone** = $\|f\| \approx 0$ (no flow at all; off-manifold, unreachable)

5.3 Relation to the existing scalar gates

The Hodge pass unifies three separate scalar computations:

Hodge component	Replaces	Used by
grad_ratio	raw ω routing score	Pass 2 routing table
harm_ratio	FK bracket k -invariant	Pass 5 HMM brackets
curl_ratio	1 – stability	Pass 3 stability table

All three are derived from one matrix decomposition: build ∂_1 once, compute $(\partial_1 \partial_1^\top)^+$ once, and all slot classifications follow.

6 The Homological Attention Gate

6.1 Standard attention and its failure

Standard self-attention computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

This is a *degree-2 computation* in the A_∞ sense: it evaluates m_2 (the bilinear Floer pairing) but ignores:

- m_0 (degree 0): does this state exist? ($W_0(u) > 0$)
- m_1 (degree 1): is this transition topologically connected?
- $m_3, m_4, \dots$: stability and Novikov corrections

A hallucination arises when the model assigns high m_2 score to a transition that is obstructed at m_0 or m_1 level.

6.2 The gated decoding rule

Definition 6.1 (Homological attention gate). *Let $\mathbf{1}_{\text{HF}_i \neq 0}(x_{t+1})$ be the topological indicator testing whether the candidate continuation x_{t+1} preserves a nonvanishing Floer class given context $X_{\leq t}$. The gated attention logits are:*

$$A_{ij} = \frac{QK^\top}{\sqrt{d_k}} + \log \mathbf{1}_{\text{HF}_i \neq 0}(j)$$

Since $\log 0 = -\infty$, the softmax assigns exactly zero probability to any token j where $\text{HF}^*(L_i, L_j) = 0$.

The gated decoding distribution is:

$$P_{\text{topo}}(x_{t+1} \mid X_{\leq t}) \propto P_\theta(x_{t+1} \mid X_{\leq t}) \cdot \mathbf{1}_{\text{HF}_i \neq 0}(x_{t+1})$$

7 The Homological Admissibility Theorem

Theorem 7.1 (Topological support restriction). *For any slot (s, h, m) with $W_0(s, h, m) = 0$ (toric obstruction, $x \notin V(I)$):*

$$P_{\text{topo}}(\text{any output} \mid s, h, m) \equiv 0$$

The support of the gated distribution satisfies:

$$\text{supp}(P_{\text{topo}}) \subseteq V(I)$$

Proof. $W_0(s, h, m) = 0 \Rightarrow \omega(s, h, m) = 0 \Rightarrow L_i \cap L_j = \emptyset \forall i, j \Rightarrow \text{CF}^* = 0 \Rightarrow \text{HF}^* = 0 \Rightarrow \mathbf{1}_{\text{HF} \neq 0} = 0 \Rightarrow P_{\text{topo}} = 0$. $\square$

Theorem 7.2 (Homological Admissibility). *Let semantic states form objects $L_i \in \mathcal{F}(M)$ in the Fukaya category of the toric symplectic manifold (M, ω) . Let decoding trajectories correspond to A_∞ morphism compositions through the Postnikov filtration $P_1 \subset \dots \subset P_5$. Any generated sequence whose continuation induces a trivial obstruction class at Postnikov stage n is excluded from decoding support. Formally:*

$$\forall x_{t+1} : \exists n \in \{1, \dots, 5\} \text{ s.t. } \text{lift}_{P_n}(x_{t+1}) \text{ obstructed} \implies P_{\text{topo}}(x_{t+1} \mid X_{\leq t}) = 0$$

Proof sketch. By induction on Postnikov stages:

- **Stage P_1 :** $x_{t+1} \notin V(I) \Rightarrow \text{embed}(x_{t+1}) = 0 \Rightarrow$ excluded by the $q \mapsto f(q)$ substitution.
- **Stage P_2 :** no demographic intersection $\Rightarrow$ absent from routing table $\Rightarrow$ score = 0.
- **Stage P_3 :** $m_3 > \tau \Rightarrow$ flagged volatile $\Rightarrow$ skipped by stability gate.
- **Stage P_4 :** m_1 propagation fails across adjacent nodes $\Rightarrow$ absent from neighbourhood table $\Rightarrow$ unreachable.
- **Stage P_5 :** $k\text{-inv} \approx 0 \Rightarrow W_0(u) \approx 0 \Rightarrow \text{HF}^* = 0 \Rightarrow \mathbf{1}_{\text{HF} \neq 0} = 0 \Rightarrow P_{\text{topo}} = 0$.

The A_∞ relation at each surgery node ensures these gates are mutually consistent. $\square$

Corollary 7.3 (Impossibility of hallucination at topological boundary). *A sequence that is topologically obstructed (lies in $V(I)^c$ or has $k\text{-inv} = 0$) cannot be generated regardless of the learned weights θ , the temperature, or the prompt.*

8 Variable-Set Semantics and Dynamic Topology

The framework generalises from static support restriction $\text{supp}(P) \subseteq \mathcal{M}$ to context-dependent admissibility:

$$\text{supp}(P_t) \subseteq \mathcal{M}_t$$

where the admissible manifold $\mathcal{M}_t$ evolves with the generation context $X_{\leq t}$.

Definition 8.1 (Sheaf of admissible universes). *In variable-set (sheaf/topos) semantics, membership is context-dependent:*

$$x \in \mathcal{M}(U) \quad \text{for contextual region } U \in \mathcal{C}$$

A token may be admissible in one discourse frame but obstructed in another. The family $\{\mathcal{M}_U\}_{U \in \mathcal{C}}$ with gluing conditions is a sheaf of admissible universes.

The AU (Arithmetic Universe) attic implements this sheaf as lazy evaluation: coproducts $L_i \sqcup L_j$ are evaluated in local context (s, h, m) via the pullback functor and are never globally materialised. Each surgery node is a stalk; restriction maps between adjacent stalks are the Floer continuation maps.

8.1 Floer continuation as context evolution

The continuation map $\Phi : \text{HF}^*(L_0, L_1) \rightarrow \text{HF}^*(L'_0, L'_1)$ under Hamiltonian deformation is the formal mechanism by which $\mathcal{M}_t$ evolves as context accumulates. A hallucination corresponds to a trajectory:

$$\gamma : [0, T] \rightarrow \mathcal{M}_t$$

that *fails to possess a nontrivial continuation class* — meaning $\Phi(\gamma) = 0$ at some step t . The decoding is terminated at that step.

9 Validated Results on the MTR Network

The framework is validated on the Hong Kong MTR network with $|V| = 81$ stations, $|E| = 89$ edges, $\beta_1 = 9$ independent cycles, 37 Markov circuits (4ti2), and 161 Graver elements.

9.1 Hodge decomposition results

The discrete Hodge pass identifies:

- **23 loop stations** ($\text{harm_ratio} > 0.05$): On one of the 9 network cycles. Highest: Olympic (0.490), Mong Kok East (0.446), Hung Hom (0.383), Austin (0.362). These receive stronger HMM bracket bounds.
- **58 tree stations** ($\text{harm_ratio} \approx 0$): On the spanning tree. Pure gradient routing. Fully live.
- **0 dead zones at peak hour**: All stations have $\omega > 0.03$ at 9am February. Dead zones emerge at off-peak hours ($\omega \rightarrow 0$).

9.2 Gate agreement

Testing all 486 slots (81 stations $\times$ 2 hours $\times$ 3 months), the Hodge gate achieves:

- **100% exact match** with the existing scalar gates (routing table + stability + k -invariant).
- **0 mismatches**: the Hodge decomposition is a mathematically equivalent reformulation of the three separate scalar computations, derived from one incidence matrix computation.

10 Discussion

10.1 What is genuinely novel

Standard transformers perform *probability maximisation* — local likelihood optimisation at each step. This architecture performs *homotopy-coherent continuation* — global trajectory validation through categorical state spaces.

The distinction is fundamental:

- Standard: *what is the most likely next token?*
- This work: *does this continuation preserve cohomological consistency across all Postnikov stages?*

10.2 Genuine vs. metaphorical contributions

We distinguish explicitly between components that are rigorously derived from the theory and those that are engineering constructs organised by the mathematical language:

Genuinely theory-derived	Engineering in mathematical language
4ti2 Markov basis computation	HNSW hierarchical indexing
Pass 1 Floer pairing embedding	EMA feedback updates
FK bracket structure	m_3 finite-difference stability
Dead-zone detection via $W_0 \approx 0$	Routing table as flat Parquet file
Hodge decomposition (one matrix op)	Threshold constants θ, τ

10.3 Known limitations

- We operate at classical limit $T = 0$. Full quantum corrections (Novikov ring) constitute Passes 6–8.
- The guarantee applies at the topological boundary. Within $V(I)$, ranking is heuristic and requires empirical validation.
- The static routing table requires recompilation when ω changes substantially. Adapton-style incremental recompilation is identified but not implemented.
- The moduli problem is computed at leading order ($\#\mathcal{M} \approx \sqrt{L_i \cdot L_j} \cdot \omega$), not from full virtual fundamental chains.

11 Conclusion

We have presented a framework in which hallucination removal is algebraic and topological rather than probabilistic. The key objects — toric ideal, Floer chain complex, Postnikov filtration, Hodge decomposition, AU sheaf — are each operationally defined and computationally instantiated on the 81-station MTR network.

The central claim is:

A hallucination is not a low-probability event. It is a trajectory that is geometrically unsupported in the admissible topology — non-liftable at some Postnikov stage. The architecture does not make such trajectories unlikely. It makes them structurally unreachable under the compiled admissibility constraints.

We are careful to distinguish geometric obstruction from semantic falsity. A blocked trajectory is *unsupported by the compiled topology*, not necessarily *logically false in all possible worlds*. The guarantee is structural constraint, not omniscience.

The strongest formal statement is the Homological Admissibility Theorem 7.2: any generated sequence whose continuation induces a trivial obstruction class at any Postnikov stage is excluded from decoding support — algebraically, not statistically.

Future work: compiler passes

The immediate algebraic programme comprises:

- **Pass 6** (Cone compiler): wall-crossing cluster mutations at interchange stations via `SyzygyProjection`.
- **Pass 7** (Vanishing cycles): seasonal Dehn twist monodromy via `ExceptionalProjection`.
- **Pass 8** (Stalk transport): Gauss-Manin connection for full m_1 parallel transport via `SpectralProjection`.

Future work: empirical validation requirements

The framework is beyond idea stage. To achieve publication, the following empirical evidence is required:

- (i) **Benchmark comparison**: baseline transformer vs. homologically-gated transformer on standard hallucination benchmarks (TruthfulQA, HaluEval, FactScore).
- (ii) **Invalid trajectory rate**: measure the fraction of generated sequences that are topologically blocked at each Postnikov stage. This quantifies how much work the gate is actually doing.
- (iii) **Long-context consistency**: demonstrate that the Floer-like continuation gate maintains coherence over longer contexts than the baseline.
- (iv) **OOD robustness**: show that the topology contributes nontrivially on out-of-distribution inputs — not merely on the training distribution.
- (v) **Ablation studies**: remove each Postnikov stage independently and measure the hallucination rate at that level. This validates the hierarchical structure.
- (vi) **Language model instantiation**: define the Lagrangians, symplectic form, and toric ideal for a concrete NLP task and validate the gate on factual QA.

Without these, reviewers will correctly characterise the system as “a symbolic validator attached to a transformer.” The experiments must prove that the topology contributes nontrivially — that the admissibility structure produces compositional generalisation and emergent long-range consistency that a standard learned mask cannot replicate.

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